

જો તું ઈચ્છે તો, આગળની સ્ટેપમાં

👉 આ જ topics માટે ચેપ્ટર-વાઈઝ important numericals,

અથવા 👉 "ટૂંકા one-liners / point-wise revision notes" પણ બનાવી આપી શકું છું.

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આ જ topics માટે ચેપ્ટર-વાઈઝ important numericals,

બરાબર tirth! હવે હું આ જ similar topics માટે ચેપ્ટર-વાઈઝ important numericals આપીશ—
ખાસ કરીને Applied Physicsની exam pattern પ્રમાણે, જ્યાંથી numericals વારંવાર પૂછાય છે.

✦ બધી numericals સીધે formula + stepwise solution હશે (Gujarati માં).

✦ તેજ topicsને જ cover કરું છું, જે તમારી બે papers માં common હતા.

✅ Chapter-wise Important Numericals (Similar Topics Only)

(100% exam-oriented – GTU pattern)

1) Micrometer Screw Gauge (Least Count & Measurements)

Numerical 1: Find Least Count

Q: એક micrometer screw gauge ની pitch = 0.5 mm છે. Circular scale માં 50 divisions છે.
Least count શોધો.

Solution:

$$\text{Least Count} = \frac{\text{Pitch}}{\text{Circular Divisions}}$$
$$LC = \frac{0.5}{50} = 0.01 \text{ mm}$$

Final Answer:

• L.C. = 0.01 mm

Numerical 2: Find diameter using micrometer

Q:

Main scale reading = 5.5 mm

Circular scale reading = 18 divisions

Least count = 0.01 mm

Zero error = +0.02 mm

Solution:

$$\begin{aligned}\text{Observed reading} &= MSR + (CSR \times LC) \\ &= 5.5 + (18 \times 0.01) \\ &= 5.5 + 0.18 = 5.68 \text{ mm}\end{aligned}$$

Zero error positive → subtract:

$$\text{True reading} = 5.68 - 0.02 = 5.66 \text{ mm}$$

Final Answer:

- Diameter = 5.66 mm
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2) Vernier Caliper Numericals

Numerical 1: Find Least Count

Vernier scale:

- 10 Vernier divisions = 9 Main scale divisions
- 1 Main scale division = 1 mm

Solution:

$$1 \text{ VSD} = \frac{9}{10} = 0.9 \text{ mm}$$

$$LC = 1 - 0.9 = 0.1 \text{ mm}$$

Final Answer:

- L.C. = 0.1 mm
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Numerical 2: Find the diameter

Given:

Main scale reading = 24 mm

Vernier coincidence = 7th division

Least Count = 0.1 mm

Solution:

$$\begin{aligned}\text{Total Reading} &= MSR + (VSR \times LC) \\ &= 24 + (7 \times 0.1) \\ &= 24 + 0.7 = 24.7 \text{ mm}\end{aligned}$$

Final Answer:

- Diameter = 24.7 mm
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3) Momentum – Law of Conservation Numericals

Numerical 1 (Very Important): Collision Problem

Q:

એક 2 kg નો પદાર્થ 5 m/s વેગથી ચાલી રહ્યો છે. તે 3 kg ના પદાર્થ સાથે અથડાય છે, જે શરૂઆતમાં સ્થિર છે.

Collision પછી બન્ને એકસાથે ચાલે છે. Final velocity શોધો.

Solution:

$$\begin{aligned}m_1 u_1 + m_2 u_2 &= (m_1 + m_2)v \\ 2(5) + 3(0) &= 5v \\ 10 &= 5v \\ v &= 2 \text{ m/s}\end{aligned}$$

Final Answer:

- Final velocity = 2 m/s
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Numerical 2: Impulse

Q:

20 N નો force 0.2 સેકન્ડ સુધી કાર્ય કરે છે. Impulse શોધો.

Solution:

$$Impulse = F \times t = 20 \times 0.2 = 4 \text{ Ns}$$

Final Answer:

• Impulse = 4 N·s

4) Surface Tension & Capillarity Numericals

Numerical 1 (Repeated in both papers):

Q:

Diameter of capillary tube = 0.07 cm

Rise of water = 4.2 cm

$\theta = 0^\circ$

Density = 1 g/cm³

$g = 981 \text{ cm/s}^2$

T (surface tension) શોધો.

Formula:

$$h = \frac{2T \cos \theta}{r \rho g}$$

Given:

$$\theta = 0 \Rightarrow \cos \theta = 1$$

$$r = \frac{0.07}{2} = 0.035 \text{ cm}$$

Solution:

$$4.2 = \frac{2T}{0.035 \times 1 \times 981}$$

Denominator:

$$0.035 \times 981 = 34.335$$

Cross-multiply:

$$4.2 \times 34.335 = 2T$$

$$144.207 = 2T$$

$$T = 72.103 \text{ dyn/cm}$$

Final Answer:

- Surface tension $\approx 72 \text{ dyn/cm}$
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5) Temperature Conversions

(Most repeated numericals – special for exam)

Numerical 1: Convert 101°F into $^\circ\text{C}$ and K

Formula:

$$^\circ\text{C} = \frac{5}{9}(^\circ\text{F} - 32)$$

$$^\circ\text{C} = \frac{5}{9}(101 - 32) = \frac{5}{9} \times 69 = 38.33^\circ\text{C}$$

$$\text{K} = \text{C} + 273 = 38.33 + 273 = 311.33\text{K}$$

Final Answer:

- $101^\circ\text{F} = 38.33^\circ\text{C}$
 - $101^\circ\text{F} = 311.33 \text{ K}$
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Numerical 2: Temperature at which Celsius = Fahrenheit

Let $C = F$

$$C = \frac{5}{9}(C - 32)$$

Solve:

$$9C = 5C - 160$$

$$4C = -160$$

$$C = -40$$

Final Answer:

- $^\circ\text{C} = ^\circ\text{F} = -40^\circ$
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6) Stress–Strain / Young's Modulus Numericals

Very Important (GTU asks frequently):

Q:

Length = 0.95 m

Radius = 8.7×10^{-3} m

Force = 72 kN = 72000 N

$Y = 2 \times 10^{11}$ N/m²

Find:

1. Stress
 2. Strain
 3. Elongation
-

Step 1: Area

$$\begin{aligned} A &= \pi r^2 = 3.14(8.7 \times 10^{-3})^2 \\ &= 3.14 \times 75.69 \times 10^{-6} \\ &= 237.66 \times 10^{-6} m^2 \\ A &= 2.38 \times 10^{-4} m^2 \end{aligned}$$

Step 2: Stress

$$\begin{aligned} \text{Stress} &= \frac{F}{A} \\ &= \frac{72000}{2.38 \times 10^{-4}} \\ &= 3.02 \times 10^8 \text{ N/m}^2 \end{aligned}$$

Step 3: Strain

$$\text{Strain} = \frac{\text{Stress}}{Y}$$

$$= \frac{3.02 \times 10^8}{2 \times 10^{11}}$$

$$= 0.00151$$

Step 4: Elongation

$$\Delta L = \text{Strain} \times L$$

$$= 0.00151 \times 0.95 = 0.00143m$$

Final Answers:

- Stress = $3.02 \times 10^8 \text{ N/m}^2$
 - Strain = 0.00151
 - Elongation = 0.00143 m
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7) Waves Numericals (Velocity formula)

Numerical:

Q:

Wavelength = 1200 cm = 12 m

Frequency = $25 \times 10^3 \text{ kHz}$ (This question has an error in units – GTU style)

Let's correct:

$$25 \times 10^3 \text{ Hz} = 25,000 \text{ Hz}$$

$$V = \lambda \times f$$

$$= 12 \times 25000$$

$$= 300000 \text{ m/s}$$

Final Answer:

- Velocity = $3 \times 10^5 \text{ m/s}$
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